

LITHODAT AI-READY DATA PIPELINE

"Clean Data First" Methodology for GDAC-SA

1 DATA INGESTION

- Multi-format adapters (PDF, CSV, databases, legacy systems)
- Automated validation against geoscience vocabularies
- Quality scoring at point of entry

Technical Lead: Moritz Theile (backend systems)
Domain Lead: Wayne Noble (validation rules, API)

2 SEMANTIC STANDARDIZATION

- SKOS-based controlled vocabularies (minerals, rock types, methods)
- Ontology mapping for heterogeneous sources
- Unit normalization (SI units, coordinate systems)

Technical Lead: Moritz Theile (system architecture)
Data Systems: Dr. Kohlmann (EarthBank ontologies)

3 RELATIONAL DATABASE STORAGE (FAIR-COMPLIANT)

- PostgreSQL with PostGIS for spatial data
- Full audit trail and provenance tracking
- RESTful APIs for programmatic access

Technical Lead: Wayne Noble (AWS infrastructure)
Quality Systems: Keith Dimech (audit compliance)

4 SGS LIVE PORTAL DEVELOPMENT

- Public-facing data portal for open geoscience data access
- Private secure area for SGS internal data and workflows
- Admin controls, user management, and role-based access
- Dual toolsets: public analytics + internal SGS-only features

Technical Lead: Wayne Noble (portal architecture)
Client Liaison: Keith Dimech (SGS requirements)
Frontend Dev: Full-stack team (React/TypeScript)

5 AI/ML APPLICATION LAYER

- Feature engineering from clean structured data
- Machine learning model training (classification, regression, clustering)
- Mineral prospectivity mapping and target generation
- Time series analysis and trend detection

ML Technical Advisor:
Dr. Behnam Sadeghi (PyImpetus, PyMiner)
AI Software Dev Lead:
Pedro Henrique Candido Ferreira (25+ years enterprise AI)